Type 1: Research Project | Phase 1 Review 2

# Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation

**Team Number: 28**

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# Project Proposal and Objectives

**Overall Project Idea:** Instead of implementing standard off-the-shelf ML algorithms, our team is proposing a **novel mathematical extension** to existing FL frameworks called the **Federated Personalized Drift-Aware Attention Framework (FPDAF)**. This custom algorithm modifies standard *FedAvg* and *CUSUM* to directly target Multivariate Healthcare Time-Series.

## Objectives to Achieve

1. **Privacy Preservation:** Train predictive models across multiple hospitals collaboratively without sharing raw patient data (Federated Learning).
2. **Personalization:** Use Local Personalization Heads to manage patient heterogeneity (Non-IID distributions).
3. **Drift Adaptation:** Implement CUSUM algorithms to detect when a patient's physiological state degrades and rapidly adapt the model locally.
4. **Clinical Trust:** Provide transparency using Temporal Attention mechanisms so doctors know *why* the model predicted a risk.

# Addressing Review 1 Panel Comments

| **Review 1 Panel Feedback** | **Adjustments Made for Review 2** |
|---|---|
| **1. Define application scope strictly** | Narrowed domain completely out of generalized data into **Healthcare ICU Time-Series (Sepsis / Deterioration Profiling)**. |
| **2. Target a well-defined dataset** | Selected the **PhysioNet Challenge 2019 Dataset** (40,000+ real multivariate ICU records). |
| **3. Articulate research novelty & Drift** | Defined the exact **CUSUM mathematical mechanism** triggering dynamic retraining when a patient's physiological state aggressively changes. |
| **4. Improve technical depth & XAI** | Designed a formal split-architecture using **Temporal Attention** so clinicians visually see *which* vital signs cause risk alerts. |
| **5. Comparative FL Discussion** | Contrasted generalized *FedAvg* with our localized *FedPer* nodes to prove handling of hospital heterogeneity. |

# Introduction and Background

**Introduction to the Problem:** In modern Intensive Care Units (ICUs), continuous monitoring systems generate vast amounts of high-frequency multivariate time-series data (Heart Rate, Blood Pressure, SpO2). Detecting critical deterioration early saves lives.

**Motivation to Solve This Specific Problem:**

- **The Data Silo Problem:** High-accuracy Deep Learning models require massive datasets, but rigid healthcare privacy laws (e.g., HIPAA) forbid combining data into a centralized server.
- **The Global Model Failure:** If multiple hospitals successfully engage in standard Federated Learning (FedAvg), the resulting "global model" fails because hospital demographics and local treatment protocols vary wildly.
- **The Black Box Problem:** Doctors cannot act on an AI alert if the model cannot explain the physiological basis for its warning.

# Research Gap Analysis

| Existing Research | Identified Limitations / Gap | Need / Why Gap Matters | Project Contribution |
|---|---|---|---|
| Centralized ML and on-device analytics systems (Papers 1, 17, 19, 23) | Do not support federated learning or privacy-preserving collaboration | Centralized data sharing violates privacy laws and is unsuitable for distributed environments | Enables collaborative analytics using federated learning without sharing raw data |
| Basic Federated Learning frameworks (FedAvg) (Papers 2, 4, 5) | Lack personalization, explainability, and analytics-level insights | A single global model performs poorly on heterogeneous client data | Introduces personalized local models with global aggregation and analytics |
| Personalized Federated Learning methods (FedPer, pFL) (Papers 3, 10, 16, 20) | Personalization exists but lacks explainability and drift awareness | Without explainability and adaptation, predictions cannot be trusted over time | Combines personalization with explainable analytics and drift handling |
| Concept driftaware FL approaches (Papers 2, 9) | Drift handling is not integrated with analytics or forecasting pipelines | Real-world data behavior changes over time, degrading prediction accuracy | Integrates drift detection with time-series analytics and forecasting |
| Federated Time-Series Forecasting works (Papers 6, 8, 1015, 18) | Primarily focus on accuracy; limited explainability and personalization | Stakeholders require transparency to trust and act on predictions | Provides explainable insights alongside federated forecasting |
| Explainable AI approaches (Papers 7, 17, 19, 23) | Explainability mostly exists in centralized or non-time-series settings | Explainability is critical for trust in sensitive domains like healthcare and energy | Adapts explainable AI techniques to federated time-series analytics |
| Healthcare-focused FL applications (Papers 9, 21, 22, 24, 25) | Application-specific solutions with limited generalization | A reusable framework is required across multiple domains | Proposes a domain-agnostic federated analytics framework |
| Security- and privacy-enhanced FL methods (Papers 13, 24, 25) | High complexity and low usability despite strong privacy guarantees | Practical systems must balance privacy, performance, and usability | Balances privacy, efficiency, and interpretability in a single framework |

# Dataset: PhysioNet Challenge 2019 (Sepsis Prediction)

**Dataset Overview:**

- **Source:** PhysioNet (Public Healthcare Dataset).
- **Data Type:** Real-world multivariate ICU patient monitoring time-series.
- **Structure:** 40,000 patient files (`.psv`), with each row representing one hourly observation.
- **Key Features:** Heart Rate, BP, SpO2, Temperature, ICULOS.
- **Target:** `SepsisLabel` (predicting deterioration).

**Why PhysioNet?**

- **Heterogeneity:** `setA` & `setB` come from different hospitals (perfect for FL).
- **Time-Series:** Sequences support LSTM & Drift tracking.
- **Clinical:** Interpretable vitals support our XAI mechanism.

# Data Preprocessing & Imputation Pipeline

**Challenge & Why it looks like a mess:** Real clinical records in PhysioNet contain massive sparsity and `NaN` gaps. High-frequency vitals (Heart Rate) are logged continuously by bedside monitors, while low-frequency lab tests (Bilirubin, Calcium) are only drawn manually by nurses every 12-24 hours.

**Engineered PyTorch Pipeline:** We developed a local Python preprocessing engine to dynamically ingest the 40,000+ '.psv' files and render clean PyTorch Tensors:

1. **Forward-Filling (Time-Series Carry):** Carries the last-known medical lab value forward in the sequence until a new blood measurement is taken.
2. **Backward-Filling & Zero-Padding:** Synthesizes initial missing conditions and replaces completely unmeasured lab variables with 0s to prevent Neural Network failure.
3. **Tensor Standardization:** Aggregates parameters via `StandardScaler` so highly varying units (e.g., Blood Pressure vs pH levels) do not skew the FPDAF gradient descent.

# Execution Plan (Components & Flow)

| Subdomain | Matching Components Identified | Execution Target |
|---|---|---|
| **Data Engineering** | Custom ICU Sequence Windows, Imputation matrices. | Preprocessing |
| **Deep Learning Base** | Extending LSTM with a **novel** Temporal Attention constraint. | Global Model |
| **Decentralization** | **Extended** *FedAvg* integrated with Attention-Aware weighting. | Federated Loop |
| **Customization** | *FedPer* baseline modified dynamically via **FPDAF** heads. | Client Local Model |
| **Statistical Math** | *CUSUM* mathematically altered to act as a Neural Network freezing layer trigger. | Drift Tracker |

**Novel Solution Pipeline:** Extract Data $\rightarrow$ Modify *FedAvg* Aggregation $\rightarrow$ Inject Attention Constraints $\rightarrow$ Sync Modified Shared Layers $\rightarrow$ Trigger Novel *CUSUM*-Adaptive Local Reset.

# Fundamentals Learned in the Semester

To execute the plan above, our team embarked on extensive upskilling over this semester:

- **Federated Learning Architecture:** Extensively reviewed *McMahan et al.'s* fundamental papers to understand gradient averaging and the shift from Centralized ML to Decentralized ML.
- **PyTorch Framework Fundamentals:** Completed MOOCs & online tutorials to transition our team's technical skills into PyTorch (specifically custom `nn.Module` networks, `DataLoaders`, and customized training loops).
- **Time-Series Analytics:** Studied LSTM sequence modeling and Attention-mechanism math (computing context vectors via softmax weights).
- **Statistical Process Control:** Learned the *CUSUM* (Cumulative Sum) statistical methodology for detecting small shifts in ongoing time-series data.

# Mathematical Foundation

## 1. Standard Federated Averaging (FedAvg Base):

$$w_{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} w_{t+1}^k \tag{1}$$

- Server aggregates weights $w_{t+1}^k$ from $K$ hospitals weighted by patient count $n_k$. (Personalization heads are strictly excluded from this sum).

## 2. CUSUM Drift Detection (Client-Side Trigger):

$$S_t = \max(0, S_{t-1} + (e_t - \mu - k)) \tag{2}$$

- $e_t$: Prediction error (residual) at time step $t$.
- $\mu$: Mean error during stable conditions.
- If the cumulative statistic $S_t > H$ (Threshold), a concept drift is flagged, triggering local model retraining.

# Algorithmic Novelty & Innovation Plan

**The Research Novelty of FPDAF:** We are not utilizing standard implementations of *FedAvg* or *CUSUM*. We are proposing a mathematical extension that fuses them into a single, responsive clinical algorithm.

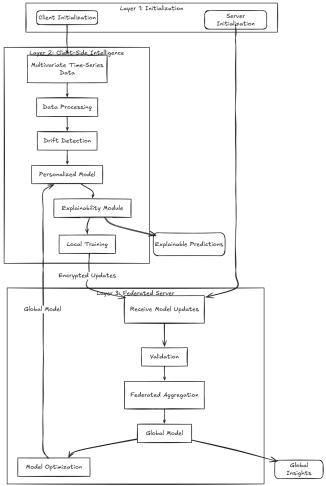
**The Novel Plan:**

- **CUSUM as a Neural Trigger:** We mathematically repurpose *CUSUM* from a simple statistical quality-control tool into an active Neural Network manipulator. When the formulation $S_t > H$ alerts, our novel code aggressively unfreezes the isolated Local Personalization Head, forcing immediate recalibration without destroying the global backbone.
- **Attention-Weighted Decentralization:** Standard *FedAvg* treats all hospitals equally based on sample size. FPDAF will introduce a novel extension aggregating weights locally, prioritized by our *Attention Mechanism*'s confidence, preventing outlier clinical anomalies from polarizing the central server.

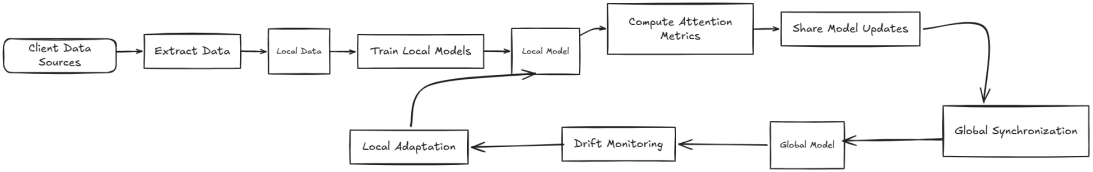
# Architecture Diagram and Explanation of Design

## Why this design?

- **Privacy:** Only encrypted gradients travel upward.
- **Local LSTM Base:** Captures generalized physiological rules.
- **Attention Layer:** Provides XAI (feature importance) right before the local prediction.
- **Personalization Head:** NEVER shared with the server. Tailors risk predictions to the specific local patient cohort.



# System Dataflow Mapping



# Summary of Works Completed (Technical Evidence)

**Phase 1 Contributions mapping to Technical Validation & Evidence:**

1. **Problem Formulation:** Successfully scoped the application domain specifically to healthcare ICU telemetry.
2. **Mathematical Formulation:** Defined formulas for exactly how CUSUM and Attention will be mathematically integrated into the *FedPer* pipeline.
3. **Architecture Design:** Finalized the multi-layered system diagram.
4. **Repository Framework Setup (Baseline Code):**
   - Initialized project GitHub repository (`https://github.com/aksharsakhi/23CSE399-Project-Phase-1`).
   - Pulled and set up the foundational **FedAvg-PyTorch base framework**.
   - Built initial custom PyTorch stubs simulating the isolated ‘LocalPersonalizedModel‘ and a synthetic ‘SimulatedICUDataset‘ proxy.

# Plans for the Oncoming Semesters (Phase 2)

**The direct roadmap to full implementation:**

| **Target Area** | **Objective for Phase 2** |
|---|---|
| **Data Preprocessing** | Implement normalization, handle missing values, and construct sequential time windows for the acquired PhysioNet dataset. |
| **Baseline Training** | Code standard *FedAvg* on PyTorch locally to establish stable forecasting baselines. |
| **FPDAF Integration** | Embed *CUSUM* threshold triggering and *Attention* layers into the personalized model heads. |
| **Federated Simulation** | Deploy 3-5 independent simulated Hospital "Clients" to test concept drift adaptation latency. |
| **Evaluation Metrics** | Benchmark system performance utilizing **Accuracy, Precision, Recall, F1-Score, and AUC-ROC**. |

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**Thank You** Questions?